07 - Text Model Interpretability

Goal: understand which words the text model uses to choose a category.

A linear SVM gives each word one weight per category. A high positive weight means the word pushes a product toward that category. We read these weights directly from the trained model.

1. Top words for each category.
2. Why the book categories get confused.
3. One wrong prediction explained word by word.

1. Train the optimised text model

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Words and word pairs: 460326
Categories: 27
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2. Top words for each category

For each category, the 8 words with the highest weight.

Code	Category	Top words
10	Used Books	edition de, and, in, der, die, cookbook, dico, und
40	Games	magideal, import, ensemble complet, la description, pal, simulator, essentials, quickio
50	Game Accessories	gaming, ps4, pour nintendo, manette, pour, skylanders, joystick, amiibo
60	Consoles	console, nom de, sony playstation, nom, pack xbox, 500 go, console de, bundle
1140	Figurines	figurine, shirt, magnet, figurines, gundam, mug, buste, watermelon
1160	Cards	carte, mtg, vf, vo, commune, pokemon, rare, attax
1180	Wargames	heroclix, warhammer, oop, miniatures, 40k, shadowrun, wargame, elfe
1280	Toys	peluche, puzzle, hélicoptère, poupée, doudou, playmobil, lego, kinder
1281	Board Games	propos, jeu, monopoly, tarot, cards, trivial, asmodee, cartes
1300	RC Models	générique, drone, maquette, 43, ho, dji, altaya, matchbox
1301	Baby Socks	auucne, baby foot, fléchettes, billard, bce, harrows, suprême, chaussettes
1302	Outdoor Toys	stream, trampoline, pêche la, nerf, balançoire, tricycle, draisienne, pêche
1320	Baby	bébé, hochet, poussette, bavoir, langer, chicco, playshoes, biberon
1560	Furniture	kitchen supplies, sweetheart93, supplies, fournitures de, sweetheart, meuble, germenly, fournitures
1920	Textiles	cdaffaires, coussin, oreiller, taie, rideaux, manique, couette, pillow
1940	Food	bio, capsules, sucre, haribo, 100g, 250g, bonbons, goût
2060	Decor	lujuik, led, decoration de, horloge, decoration, décor germenly, housekeeping, décoration
2220	Pets	pour chien, pet, chien, pour chat, laisse, chiens, zolux, chiot
2280	Magazines	du 01, 06, 04, 08, magazine, du 15, 03, hors série
2403	Book Lots	tomes, lot, volumes, livres, revues, numéros, éditions, interview
2462	Used Games	jeux, ps2, pc, vtech, xbox, ps3, wii, jeu jaquette

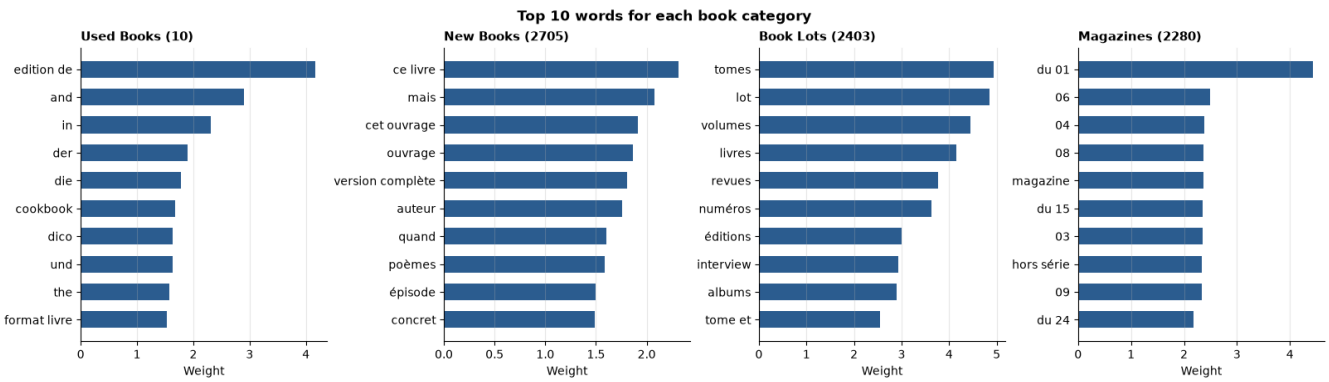
Code	Category	Top words
2522	Stationery	lindner, stylo, a4, agenda, clairefontaine, marque page, enveloppes, encre
2582	Garden Furniture	de jardin, okadaopa, de habitat, kitchen, jardin, terrasse, stainless, parasol
2583	Pool	piscine, spa, intex, sauna, piscines, pompe, cellule, bestway
2585	Garden Tools	tools, arrosage, tondeuse, outil, tronçonneuse, lame, gardena, jardinage
2705	New Books	ce livre, mais, cet ouvrage, ouvrage, version complète, auteur, quand, poèmes
2905	PC Games	en téléchargement, téléchargement, jeu en, forts de, points forts, forts, points, configuration requise

Analysis

- Most categories have clear, sensible words: *piscine* and *intex* for Pool, *chien* and *laisse* for Pets, *warhammer* and *heroclix* for Wargames, *bébé* and *poussette* for Baby.
- Some top words are brand or seller names (*sweetheart93*, *germenly*, *lujuik*, *cdaffaires*). The model learned the sellers, not only the products. This works on this data but may not hold for new sellers.
- Some categories rely on the style of the text, not the product: New Books uses *ce livre*, *cet ouvrage*, *auteur* (words from a book summary), and Used Books uses English and German words (*and*, *der*, *die*) because many used books have foreign titles.

3. Why the book categories get confused

The four book categories, side by side. Each bar is the weight of one word for that category.



Word	Used Books	New Books	Book Lots	Magazines
livre	0.04	0.16	1.97	-1.43
tome	0.77	0.51	1.34	-1.10
edition	0.03	-0.99	0.19	-0.55
édition	0.70	-0.50	0.79	-1.16
roman	0.24	0.49	0.58	0.44
auteur	-0.18	1.75	-0.45	-0.64
collection	0.30	-1.20	1.00	-0.48

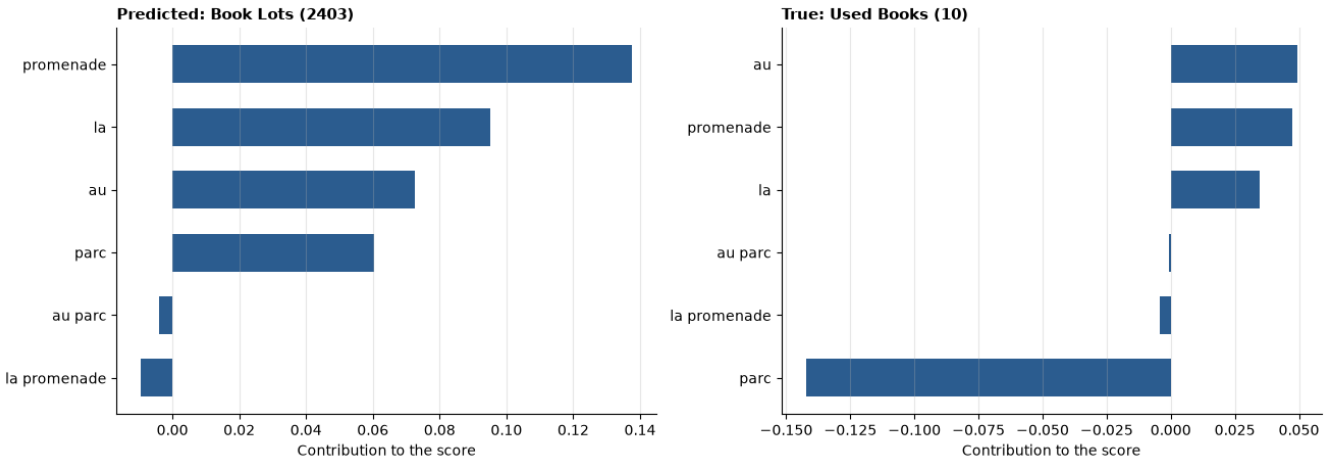
Analysis

- Each book category has its own strong words: *tomes*, *lot*, *volumes* for Book Lots, dates like *du 01* for Magazines, summary words for New Books, foreign words for Used Books.
- The common book words are weak and split across categories. *Roman* has almost the same weight (0.24 to 0.58) in all four. *Tome* and *édition* point to both Used Books and Book Lots.
- So a book title without one of the strong words (no "lot", no date, no summary) gives the model almost nothing to decide with. This is why book categories make 7 of the top 10 confused pairs.

4. One wrong prediction explained

We take a used book (10) that the model predicted as a book lot (2403). For each word in the product, its contribution to a category is its TF-IDF value times its weight. We compare the words that pushed toward the wrong category and toward the true one.

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Title: la promenade au parc
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Score for Book Lots (2403): −0.50
Score for Used Books (10):  −0.68
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Analysis

- The product is a used book with only a short title: *la promenade au parc*. It has no description.
- No word in it is a book word. The model only sees *promenade*, *la*, *au* and *parc*.
- Both scores are negative (-0.50 for Book Lots, -0.68 for Used Books), so the model is not confident in any category. Book Lots wins by a small margin, mainly because of *promenade*. *Parc* pushes away from Used Books.
- This shows a limit of text models: when the text is short and generic, there is not enough information. The image can help in these cases (see notebook 06).

Conclusion

- The text model relies on clear, specific words, which explains its good overall score.
- It also learned seller names and writing style, which may not hold for new sellers.
- Book categories share weak, common words. Short book titles without a strong word are the main source of errors.